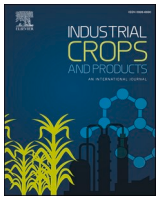
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Lignin dissolution and lignocellulose pretreatment by carboxylic acid based deep eutectic solvents

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ABSTRACT

Deep eutectic solvents (DESs) composed by carboxylic acids and quaternary ammonium salts are potential candidate for lignocellulose dissolution and pretreatment. To both explore the most efficient DESs for lignocellulose dissolution and pretreatment as well as unveil the process mechanism, DESs composed by carboxylic acids as the hydrogen bond donors (HBDs) and quaternary ammonium salts as the hydrogen bond acceptors (HBAs) were prepared and further applied for kraft lignin (KL) dissolution as well as poplar and pine pretreatment. We found DES composed by lactic acid (LA) and allyltrimethylammonium chloride (ATMAC) with mole ratio of 2:1 is high-efficient for the dissolution of KL. In addition, 70% poplar lignin and 82% pine lignin could be removed at 120 ℃ for 6 h with a solid-to-liquid ratio of 1:10. Furthermore, the -OH and -COOH groups in HBDs = of DESs provided active protons to cleave ether bonds and part of carbon-carbon bonds during lignin dissolution and extraction. Additionally, the KL solubility and the Kamlet-Taft (K-T) parameters (α and π *) of the DESs showed good positive correlation. These findings give a deep insight into the dissolution mechanism of KL and pretreatment of lignocellulose, which is helpful for designing robust DESs for dissolution and pretreatment of lignocellulose.

1. Introduction

Lignin is the most abundant aromatic biopolymer in nature. Meanwhile, it is a sustainable precursor for producing aromatic chemicals. In the past decades, transformation of lignin into functional materials and value-added chemicals has drawn numerous attentions. Generally, dissolution is a prerequisite for effectively utilization/conversion of lignin (Chen et al., 2020b; Ragauskas Arthur et al., 2014; Sun et al., 2018; Zhang et al., 2017a, 2017b). However, lignin is composed by the cross link of three phenylpropane structure units through diverse linkages, which makes the dissolution of lignin highly difficult (Hong et al., 2020b; Li et al., 2015; Liu et al., 2017).

Until now, several types of solvent systems have been applied in the dissolution of lignin, for example, ionic liquids (ILs), (An et al., 2015; Ji et al., 2012; Liu et al., 2019b; Rashid et al., 2016) mixture of ILs and water, (Xu et al., 2017) mixture of water and organic solvents (Zhang et al., 2017a, 2017b) or γ-valerolactone (Xue et al., 2016), and liquid ammonia (Strassberger et al., 2015). Although these solvent systems could dissolve lignin, they have some obvious drawbacks. Organic solvents are volatile, while liquid ammonia requires high pressure and low temperature (Strassberger et al., 2015). ILs are usually expensive and viscous (Cao et al., 2012; Wang et al., 2017a, 2017b). It is still a challenge for designing novel and robust solvent systems to dissolve lignin.

Deep eutectic solvents (DESs) are cheap, easy to prepare, and can be designed to be non-toxic, biocompatible and biodegradable (Abbott et al., 2004, 2003; Paiva et al., 2014; van Osch et al., 2020; Smith et al., 2014; Yu et al., 2021). Recently, applications of DESs in pretreatment of lignocellulose have gained significant attention (Francisco et al., 2012; Lou and Zhang, 2022; Wang and Lee, 2021; Xu et al., 2021). Reports have shown that organic acids based DESs are highly efficient for lignocellulose pretreatment (Tan et al., 2019; Zhang et al., 2016; Francisco et al., 2012). For example, Alvarez-Vasco et al. used DESs composed by organic carboxylic acid (*i.e.*, acetic acid, levulinic acid, and lactic acid (LA)) and choline chloride for delignification of hardwood and softwoods, and LA based DES showed the highest lignin extraction yields (78.5% from hardwoods and 58.2% from softwoods) (Alvarez-Vasco et al., 2016). Lynam et al. found that LA-based DESs could dissolve alkali lignin with the highest solubility of 13 wt% (Lynam et al., 2017). Hong et al. reported that LA and choline chloride-based DES is more efficient on separating alkali lignin from poplar than oxalic acid

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and choline chloride-based DES (Hong et al., 2020a). Liang et al. used LA-based DESs for pretreatment of corn stover and found that both alkyl chain length and functional groups of HBAs significantly influence the efficiency of pretreatment (Liang et al., 2021). These discussions indicated that carboxylic-based DESs (especially LA) are efficient for lignocellulose pretreatment and dissolution because of the unique molecular structure of LA, in which there are simultaneously one carboxyl group and one hydroxyl group. However, the solubility of lignin in the currently developed LA-based DESs was not satisfactory, and the mechanism of the better performance of LA-based DESs has not been well unveiled.

To enhance the solubility of lignin in LA-based DESs and reveal the enhancing effect of LA on lignin dissolution, herein, several carboxylic acids (*i.e.*, LA, malic acid (MA) and glycolic acid (GA)) and quaternary ammonium salts (allyltrimethylammonium chloride (ATMAC), choline chloride (ChCl), tetramethylammonium chloride (TMAC), and tetraethylammonium chloride (TEAC)) based DESs were prepared (Fig. S1). The solubility of kraft lignin (KL) in these DESs at 50 ℃, 60 ℃ and 70 ℃ were measured. The relationship between lignin solubility and the Kamlet-Taft (K-T) parameters of the DESs was analyzed. The structural changes between the original and the regenerated lignin were characterized. In addition, pretreatment of poplar and pine with the designed DESs was conducted and the enhanced effect of the LA-based DESs on KL dissolution and lignocellulose pretreatment was investigated.

2. Materials and methods

2.1. Materials

Kraft lignin (KL, with low sulfonate content) was purchased from Sigma-Aldrich. Poplar (5 years old) wood was obtained from Hebei Province and pine (Pinus yunnanensis, 6 years old) wood was obtained from Yunnan Province. Choline chloride (ChCl), lactic acid (LA), allyltrimethylammonium chloride (ATMAC), and glycolic acid (GA) were obtained from Shanghai Aladdin Biochemical Co., Ltd (Shanghai, China). Malic acid (MA) and tetramethylammonium chloride (TMAC) were bought from Beijing Innochem Technology Co., Ltd (Beijing, China). Tetraethylammonium chloride (TEAC) was purchased from Energy Chemical (Beijing, China).

2.2. Preparation of DESs

The DESs were prepared by mixing the HBDs and HBAs at a molar ratio of 1:1 or 2:1 in a glass vial. Subsequently, the mixture was heated to 70 ℃ with a mechanical stirrer at 500 rpm until a transparent liquid was obtained. Finally, all DESs were stored under vacuum at 60 ℃ for 24 h to remove the contaminated trace water.

2.3. Determination of KL solubility in the prepared DESs

The solubility of KL was measured based on our previously reported route (Yu et al., 2020). The KL solubility measurement consisted of a total of two steps. The first step was roughly measured, and 30 mg of KL was added to *ca.*1 g DES each time, and the addition is repeated until the added KL could not be completely dissolved within 1 h. The second step was accurately measured. 90% amount of the first-step achieved KL solubility was added to *ca.*1 g DES, then 3 mg of KL was added to DES each time until the KL could not be dissolved in 1 h. The detailed information was presented in Supplementary Information section.

2.4. Pretreatment with DESs and composition analysis

The poplar and pine particles of sizes between 40 and 60 mesh were obtained by smash and then sieve the poplar and pine sample. After that, the wood powder was extracted with toluene–ethanol (2:1, v/v) in a Soxhlet instrument for 6 h. The poplar/pine powder were mixed with LA-ATMAC-2:1 or LA-ChCl-2:1 with a mass ratio of 1:10 and put in an oil bath at 120 ℃ for 6 h, and followed by dilution with ethanol. The solid components were further separated and washed with ethanol and deionized water until the filtrate clarifies, and then freeze-dried for the followed analysis. The detailed pretreatment process was given in Supplementary Information section. The chemical compositions of poplar sample and the solid substrates were determined according to a reported procedure (Sluiter et al., 2010). The content of acid-insoluble lignin (AIL, Klason lignin) was determined by measuring the weight of the residue after drying. The absorbance of acid-soluble lignin (ASL) at 240 nm was determined by UV2310-vis spectrophotometer. The determination of monomeric sugars in the aqueous soluble fraction performed on HPAEC system (Dionex ICS 3000, USA) equipped with a Carbopac TM PA-20 column (3 mm × 150 mm, Dionex, Sunnyvale, USA) and an amperometric detector. The details could be seen in Supplementary Information section too.

2.5. Structural characterization of KL

Nuclear magnetic resonance (NMR) spectra were performed on a Bruker Avance 400 MHz spectrometer. The weight-average (M_w) and number-average (M_n) molecular weights of the two acetylated KL samples were analyzed by Gel Permeation Chromatography (GPC, Agilent 1200). Scanning Electron Microscope (SEM) analysis was conducted on a Hitachi S-4800 Scanning Electron Microscope operated at 15 kV. Thermogravimetric analysis (TGA) experiment was conducted on a TGA Q500 (TA Instrument Company, America). Fourier Transform infrared (FT-IR) spectra were recorded on a Prestige-21 FT-IR Spectroscopy (Shimadzu, Japan). Detailed information was given in Supplementary Information section.

3. Results and discussion

3.1. Dissolution of KL in DESs

The prepared DESs (Table S1) were denoted as HBD-HBA-x, where x represents the molar ratio of HBD and HBA. In Table S1, LA-TMAC-1:1 and GA-TMAC-1:1 could not form DESs. Subsequently, the prepared DESs were applied to dissolve KL at different temperatures (Fig. 1).

Fig. 1 shows that the solubility of KL in all examined DESs increases with the dissolution temperature. When the molar ratio of HBD and HBA was 1:1, at 50 ℃, LA-ATMAC-1:1 and LA-ChCl-1:1 could dissolve KL, but the solubility of KL in LA-TEMC-1:1 is less than 1 wt%. As shown in Fig. 1, LA-ATMAC-1:1 provides the best lignin solubility of 15.6 wt% at 70 ℃, while the lignin solubility is only 2.9 wt% in LA-TEMC-1:1 at 70 ℃. Fig. 1B shows that the solubility of KL is significantly enhanced to 22.5 wt% when the molar ratio of HBD and HBA increase to 2:1 for LA-ATMAC-2:1 at 70 ℃. Notably, the solubility of KL at 70 ℃ increases with the following order: LA-TEAC-2:1 < LA-TMAC-2:1 < LA-ChCl-2:1 < LA-ATMAC-2:1. For the GA-based DESs, a similar tendency was observed (Fig. 1C and D). The GA-ATMAC-x showed the best performance at the same GA/HBA molar ratio, and high molar ratio of GA and HBAs is beneficial for the dissolution of KL. Meanwhile, the solubility of KL at 70 ℃ increases with the following order: GA-TEAC-2:1 < GA-TMAC-2:1 < GA-ChCl-2:1 < GA-ATMAC-2:1. GA-ATMAC-2:1 shows the maximum KL solubility of 19.1 wt% at 70 ℃ among all GA-based DESs (Fig. 1). Table S2 indicates that all the prepared MA-based DESs show poor performance on the KL dissolution even at 70 ℃. From the above results, the functional groups in the employed HBAs significantly affect the KL dissolution, which followed the order: C═C > -OH > alkyl, and meanwhile, the KL solubility decreases with the increase of the cation size in HBAs, *e.g.*, LA-TEAC-2:1 < LA-TMAC-2:1 and GA-TEAC-2:1 < GA-TMAC-2:1.

Besides, for the DESs with same HBA, LA and GA-based DESs show better performance on KL dissolution at all determined temperatures (*i.e.*, 50 ℃, 60 ℃ and 70 ℃) (Fig. 1), while the solubility of KL in MA-

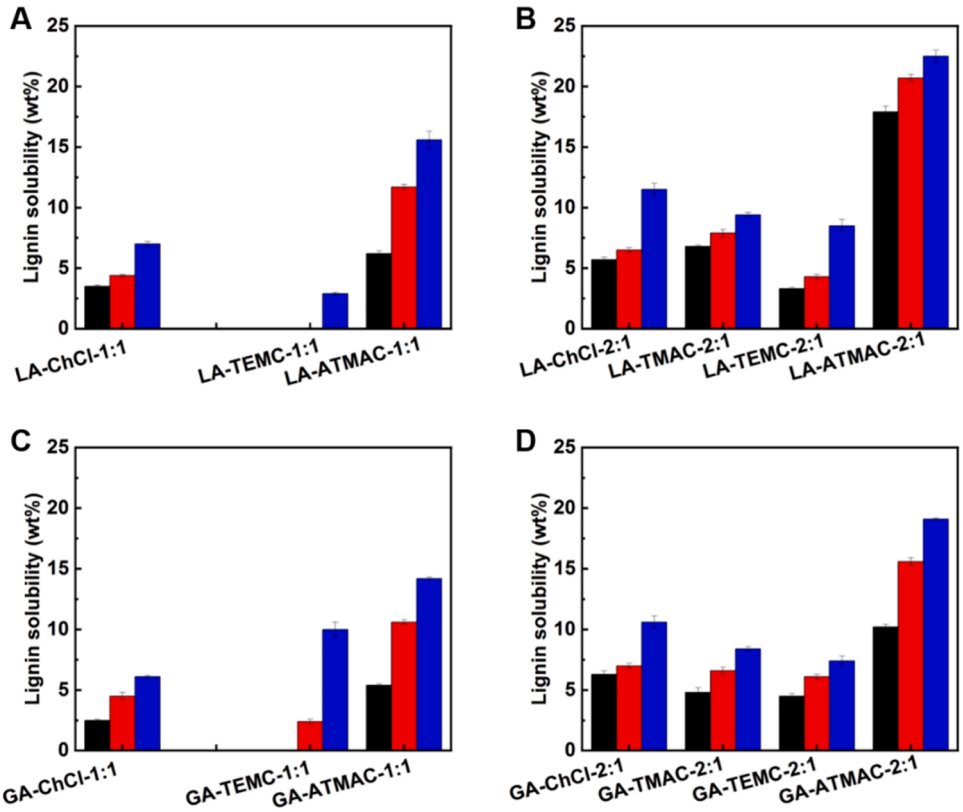


Fig. 1. The solubility of KL in the prepared DESs. The dark, red, and blue bars represented the results at 50 ℃, 60 ℃ and 70 ℃, respectively.

based DES is very low. From the chemical structure of the used HBDs (LA, MA, and GA), the main difference of the used HBDs lies in that there are two -COOH groups in MA, while only one -COOH group in both LA and GA. Meanwhile, all the three HBDs contain one -OH group. The -OH group and -COOH groups in the HBD are beneficial for lignin dissolution, but more than one -COOH groups weaken the lignin dissolution (Tan et al., 2019). Consistent with Francisco's results (Francisco et al., 2012), MA-based DESs show poor KL dissolution ability. The presence of two -COOH groups in MA increases the number of hydrogen bonds and extensive dimer chains, which significantly increase the viscosity (Table S3) and thus decrease mass transfer and KL dissolution. In comparison, GA- and LA-based DESs perform similarly on KL dissolution, and the solubility of KL in these two types of DESs vary with the type of HBA and the HBD/HBA molar ratio. For example, the lignin solubility in LA-TMAC-2:1 is higher than that in GA-TMAC-2:1, while GA-TEAC-2:1 shows higher solubility than LA-TEAC-2:1. Moreover, LA-ChCl-1:1 shows higher KL solubility than GA-ChCl-1:1, but the solubility of KL in LA-ChCl-2:1 is lower than that in GA-ChCl-2:1. The different performance of the prepared LA- and GA-based DESs to dissolve KL will be discussed in the following section.

3.2. The effect of DESs' K-T parameters on KL dissolution

K-T empirical parameters (*i.e.*, α, β, and π *) of the solvents could be employed to evaluate the ability of the solvents to dissolve lignocellulose components. These parameters can be applied to quantify hydrogen-bond donating ability (acidity, α), hydrogen-bond accepting ability (basicity, β) and the polarizability (π *) of the solvents. The K-T empirical parameters of the prepared LA- and GA-based DESs were determined and given in Table S4. The correlation between KL solubilities (Fig. 1) and the K-T parameters Table S4 is very poor if all the LA- and GA-based DESs are considered. However, after carefully analyzing the obtained results, some regularities could still be observed.

First, the solubilities of KL in LA- and GA-based DESs are similar when they have the same HBA and molar ratio of HBD/HBA (Fig. 1). Correspondingly, K-T empirical parameters of LA- and GA-based DESs with same HBA and molar ratio of HBD/HBA are similar (Table S4) although the correlation between KL solubility and the K-T parameters is not completely consistent.

Second, compared with other HBAs, the alkyl chain of TEMC is longer and its volume is larger than that of the other three HBAs, which leads to its larger steric hindrance, causing the HBDs has a weak ability to form H-bonds with it (Zhou et al., 2022). In addition, the molar ratio of HBAs and HBDs also affects the dissolution ability of lignin. When the molar ratio of HBDs increased, LA and GA could provide more active protons, which promoted the dissolution of lignin (Zhou et al., 2022). As a result, LA-TEAC-1:1 and GA-TEAC-1:1 show the lowest values of α and π * , indicating that they have the poorest hydrogen-bond donating ability and lowest polarity. In contrast, the solvents possessing stronger hydrogen-bond donating ability and higher polarity have better ability to dissolve lignin (Liu et al., 2019a). Thereby, the solubility of KL in LA-TEAC-1:1 and GA-TEAC-1:1 is very low.

Third, the correlation between KL solubility in LA- and GA-based DESs and the K-T parameters was further analyzed by the statistical method of linear solvation energy relationship (LSER) fitted by multiple linear regression (Ab Rani et al., 2011; Florindo et al., 2018; Gilbert et al., 2019). It was observed that the KL solubility has a good positive correlation with the values of α and π * (Table 1 and Fig. 2). However, although DESs had a high β value, the correlation with the β value is not good. It could be seen from Table 1 that the correlation between KL solubility of LA-based DESs at 70 ℃ and the K-T parameters follows the equation $\ln S = -30.44 + 24.79\alpha$ and $\ln S = -29.28 + 30.30\pi^*$. The correlation between KL solubility of GA-based DESs and the K-T parameters follows the equation $\ln S = -87.68 + 67.03\alpha$ and $\ln S = -84.53 + 81.90\pi^*$ at 50 ℃, and $\ln S = -97.41 + 74.74\alpha$ and $\ln S = -93.88 + 91.30\pi^*$ at 70 ℃. On the contrary, the correlation between KL solubility of LA-based DESs at 70 ℃ and the K-T parameters follows the equation $\ln S = 0.92 + 2.79\beta$. The K-T parameters follows the

Table 1
Results of Kamlet-Taft LSER Fits for the dissolution of KL in DESs.

DESs	Molar ratio	Temperature (℃)	Parameters	*p* values[a]
LA-based DESs	2:1	70	α	$XYZ_0 = 0.042$, $\alpha = 0.039$
		70	π*	$XYZ_0 = 0.045$, $\pi^* = 0.042$
		70	β	$XYZ_0 = 0.901$, $\beta = 0.823$
GA-based DESs	2:1	50	α	$XYZ_0 = 0.009$, $\alpha = 0.009$
		50	π*	$XYZ_0 = 0.010$, $\pi^* = 0.010$
		50	β	$XYZ_0 = 0.330$, $\beta = 0.160$
		70	α	$XYZ_0 = 0.023$, $\alpha = 0.022$
		70	π*	$XYZ_0 = 0.024$, $\pi^* = 0.023$
		70	β	$XYZ_0 = 0.225$, $\beta = 0.191$

[a] In this work, a maximum p value was < 0.05, if values proved statistically significant ($p < 0.05$), the correlation was accepted. Entry1 and 2, $p > 0.05$, there was no correlation between AL solubility and the Kamlet-Taft parameters.

equation $\ln S = 0.68 + 2.35\beta$ and $\ln S = 1.13 + 2.55\beta$ for GA-based DESs at 50 and 70 ℃. All *p* values were greater than 0.05. Therefore, there was no linear correlation between lignin solubility and β value (Table 1).

From the above experimental results and discussions, we could deduce that the DESs with relative higher values of α and π * generally show better performance on lignin dissolution. This is because, the high α values of DESs were key factor that correlate with the lignin extraction and dissolution efficiency during biomass pretreatment. DESs with a high α value could easily donate H-bonds to release H^+, hence a higher degree of lignin dissolution (Lee et al., 2021; van Osch et al., 2017; Xia et al., 2018). Although there was literature showing that both high α and β of DESs were effective for lignin extraction, however, in this paper, there was no linear correlation between lignin solubility and β value (Lee et al., 2021; van Osch et al., 2017). Moreover, the values of α and π * could be an indicator for DESs to dissolve lignin. Besides the ability to donate H-bonds and the polarizability, the viscosity of the DESs influences the lignin dissolution, and lower viscosity benefits KL dissolution.

In addition, the DESs composed by LA and ATMAC are more efficient for KL dissolution than previous reported LA-based DESs (Table S5) (Francisco et al., 2012; Lynam et al., 2017). For example, the KL solubility in LA-ATMAC-2:1 could reach 20.7 wt% at 60 ℃ (Table S5, entry 1) and 15.6 wt% in GA-ATMAC-2:1 (Table S5, entry 3). In comparison, the solubilities of KL in LA-histidine-9:1 is only 11.88 wt% at 60 ℃ (Table S5, entry 9) (Francisco et al., 2012). Moreover, the highest solubility of KL in LA-based DESs reported by Lynam et al. is only 13 wt% (Table S5, entries 10–12) (Lynam et al., 2017). These results confirm the good ability of LA-ATMAC-2:1 to dissolve KL.

3.3. Characterization of the original and regenerated KL

The dissolved KL could be easily regenerated from LA-ATMAC-2:1 using water as anti-solvent. To determine the potential structure changes, the original and regenerated KL were characterized by ^{1}H NMR spectra, 2D HSQC NMR spectra, FT-IR spectra, SEM, and TGA.

^{1}H NMR spectra were firstly employed to examine the chemical structure of original KL and the regenerated one (Fig. 3). Compared with the spectra of original KL, a new sharp peak, corresponding to the Hβ of resinol substructures, (An et al., 2015) appears at 3.05 ppm in the ^{1}H NMR spectra of the regenerated KL, which indicates the formation of resinol substructures with the β-β′ linkage *via* recondensation reactions. Meanwhile, another two new peaks at 6.1 and 5.6 ppm are observed, which are assigned to Hα of β-O-4 and β-5 structure (Rashid et al., 2016). These results clearly confirm that the chemical structure of KL has been changed during dissolution and regeneration. Moreover, after the dissolved KL was regenerated, the weak peak at 9.44 ppm disappeared (Fig. 3B), suggesting that chemical reactions in active formyl group occurred, which is consistent with the reported results (An et al., 2015). In addition, although there are some differences for the signals of methoxyl groups (intense peak at 3.79–4.07 ppm) between the original KL and the regenerated one, the changes in the methoxyl contents could not be accurately assessed owing to the signal (3.3 ppm) of residual water in DMSO-d_6 and the used KL (An et al., 2015).

2D NMR is a very useful technique to analyze the structure of lignin (Mansfield et al., 2012). The 2D HSQC NMR spectra of the original KL and the regenerated one are subsequently provided in Fig. 4. The signals in 2D HSQC NMR spectra are assigned based on the previous results (Balakshin et al., 2003; Fernández-Costas et al., 2014; Kim et al., 2008; Wen et al., 2014). The detail peak assignments and the main substructures in KL are provided in Table S6 and Fig. 4E, respectively. First, the aliphatic region showed different linkages in KL (Fig. 4 A and 4 C), including β-O-4′, β-5′, and β-β', *etc*. The intense signal at 55.2/3.71 ppm, attributed to methoxyl groups, are observed in spectra of original KL and the regenerated one. The signals at 70.8/4.75, 84.3/4.25 and 59.9/3.38, 3.57 ppm represent the C-H correlations at the α, β, and γ positions of the β-O-4′ linkage (A), respectively. By comparing the spectra in Fig. 4 A and 4 C, it is obvious that the content of β-O-4′ structure (α-γ) decreases after dissolution and regeneration. Moreover, a new signal at 51.6/3.02 ppm (B_β) is observed in the spectra of the regenerated KL (Fig. 4 C), indicating the breakage of the B_α bond after the KL was dissolved and regenerated. This result is consistent with the appearance of new peak at 1734 cm^{-1} in FT-IR spectra (Fig. S4). The presence of B_α (at 84.3/4.59 ppm) and B_γ (at 70.5/3.73, 3.96 ppm) in the original KL and B_α, B_β and B_γ in the generated KL represents the resinol substructures (B). The signals at 86.2/5.45 ppm (C_α-H_α), 51.3/3.64 ppm (C_β-H_β) and 62.8/3.67 ppm (C_γ-H_γ) confirm the existence of phenylcoumaran substructures (C) in both KL. The content of β-O-4′ aryl ether in the original KL is 65.20/100Ar, while it drops sharply to 11.70/100Ar in regenerated KL (Figs. 4B and 4D). It implies that the dissolution and regeneration of KL from the DES (LA-ATMAC-2:1) system caused the reduction of β-O-4′ content. Meanwhile, the contents of other carbon-carbon bonds (such as β-5′ and β-β') are reduced after being regenerated (Figs. 4B and 4D). As shown in Figs. 4B and 4D, the chemical shift of the guaiac-based unit changed from 6.69 to 6.97 ppm to 6.73–6.99 ppm after being regenerated, which is caused by the increase of the electron cloud density in the guaiac-based unit and the occurrence of a condensation reaction to generate new carbon-carbon bonds (Shen et al., 2019).

The surface structure of original KL and the regenerated one were characterized by scanning electron microscope (SEM). As shown in Fig. S2, the surface structure of the original KL was relatively uniform and smooth, while the regenerated KL was rougher, indicating that the physical structure of lignin changed during the dissolution and regeneration. Moreover, the molecular weight (M_w and M_n) significantly reduced after regeneration measured by gel permeation chromatography (Table 2), implying that the regenerated KL was dissociated or depolymerized in comparison with the original KL. Meanwhile, polydispersity index (M_w/M_n) of regenerated KL decreased, indicating that the dispersion of the KL became better. Additionally, in TG experiments (Fig. S3), the weight of both KLs decreased rapidly at the temperature range of 180–400 °C because the cleavage of ether bonds and the evaporation of monomeric phenols (Faravelli et al., 2010). More importantly, the decomposition temperature for the regenerated KL was lower than that of the original KL. Meanwhile, the regenerated KL provided less residual (37.32 wt%) than the original KL (54.85 wt%), suggesting that the regenerated KL occurred chemical changes in the process of dissolution and regeneration. Through these characterizations, it was proved that the surface structure of KL changed during the process of KL dissolution and regeneration. Combined with the results of

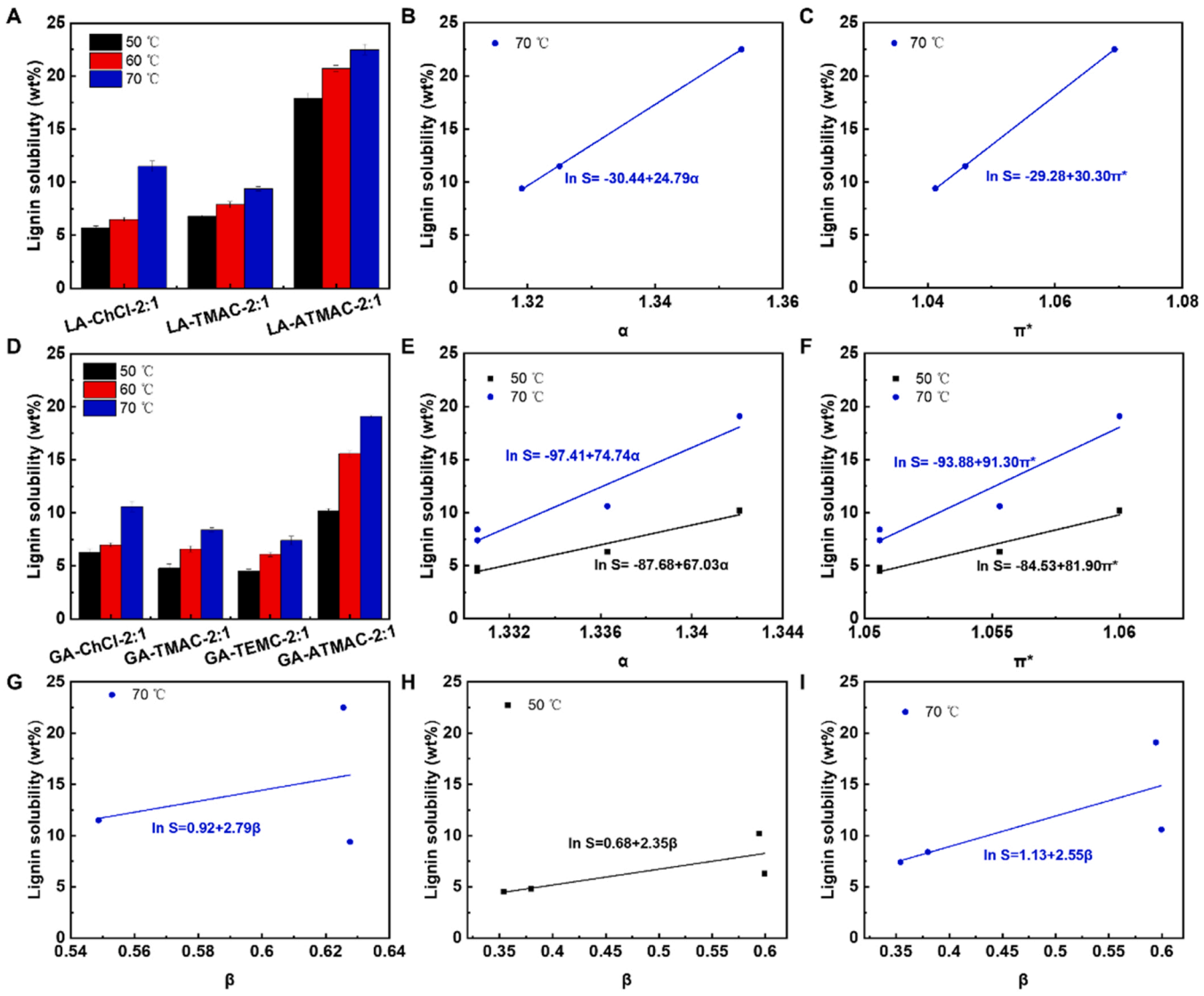


Fig. 2. (A) The solubility of KL in LA-based DESs. The dark, red, and blue bars represent the experiments conducted at 50 ℃, 60 ℃ and 70 ℃, respectively. (B) and (C) The correlation between the solubility of KL and the α and π* values of LA-based DESs. (D) The solubility of KL in GA-based DESs. (E) and (F) The correlation between the solubility of KL and the α and π* values of GA-based DESs. (G), (H) and (I) The correlation between the solubility of KL and the β values of LA-based DESs and GA-based DESs.

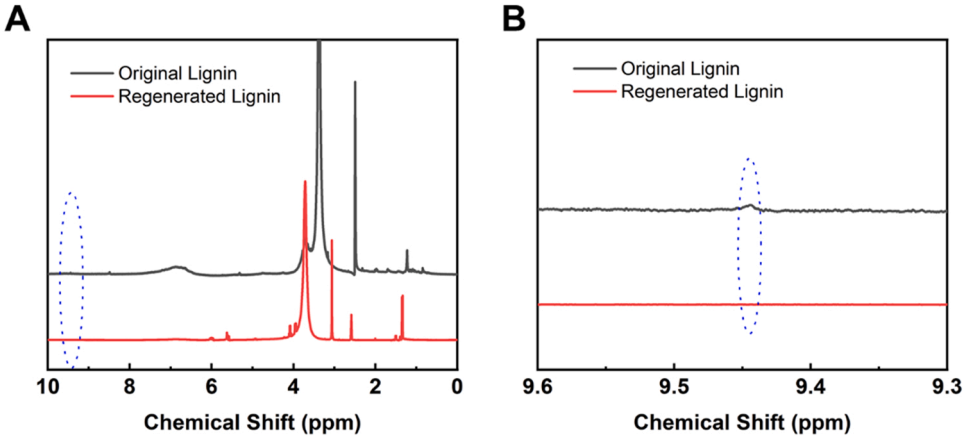


Fig. 3. ^{1}H NMR spectra of the original KL and the regenerated one.

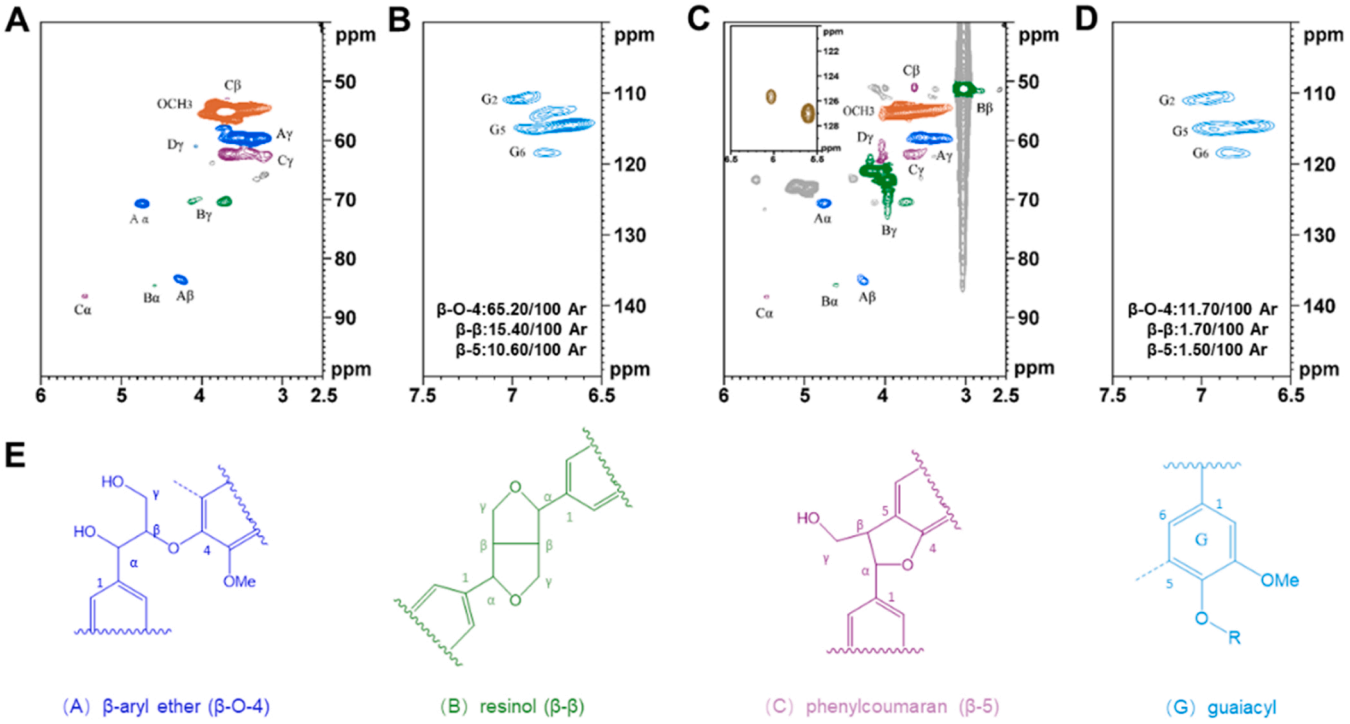


Fig. 4. 2D HSQC NMR spectra of the original KL (A and B) and the regenerated one (C and D): The side chain regions (A and C), and the aromatic regions (B and D). Main substructures of lignin, including side chain linkages and aromatic units (E).

Table 2
Molecular weights (M_w and M_n) and polydispersity index (M_w/M_n) of the original KL and the regenerated one.

Sample	M_w (g/mol)	M_n (g/mol)	Polydispersity index
Original KL	2441	1450	1.68
Regenerated KL	1315	1087	1.21

GPC and TGA, it showed that KL with small molecular weight dissolved preferentially in this process. In addition, the reduced polydispersity index of KL indicated that the lignin molecular weight distribution was relatively uniform after KL regeneration and offered a great potential in high-value applications of lignin in chemical industry (Wang et al., 2017a, 2017b).

Finally, FT-IR spectra were employed to characterize the original and the regenerated KL (Fig. S4). Characteristic peaks of both original and regenerated KL were observed, and which are consistent with previous works (Xu et al., 2017). Some of the characteristic peaks are assigned as follows. Peak at 3434 cm^{-1} is assigned to the stretching vibration of phenolic and aliphatic hydroxyl groups. Peaks at 2936 and 2837 cm^{-1} could be assigned to the tensile C-H vibrations in CH_3/CH_2 and OCH_3 groups. Peaks at 1600, 1508 and 1419 cm^{-1} are assigned to aromatic ring stretching vibrations. Peak at 1458 cm^{-1} is the in-plane asymmetric deformation vibration of C-H in CH_3 and CH_2. Peak at 1266 cm^{-1} is the stretching vibration of C-O in the guaiacyl type. Peak at 1215 cm^{-1} is the stretching vibration of C-O (H) and C-O (Ar) in the phenolic hydroxyl and ether. And peak at 858 cm^{-1} is out-of-plane deformation vibration of aromatic C-H in the guaiacyl type. A new band at 1734 cm^{-1} in the spectra of the regenerated KL indicates the breaking of the B_α bond and the damage of the C_α double bond (Chen et al., 2019), which was consistent with the results of 2D HSQC NMR spectra. In general, the results of NMR and FT-IR spectra showed that ether bond, part of carbon-carbon bond and Bα bond were broken during the KL dissolution, and the regenerated KL occurred chemical changes in the process of dissolution and regeneration.

3.4. Pretreatment of poplar and pine by DESs

After examined the solubility of KL, we further explored the pretreatment performance of the DESs on poplar and pine, which are reprehensive of hardwood and softwood species, respectively. The chemical composition of samples (wt%) were determined by standard procedures and the results are presented in Table 3. The pretreatment experiments were carried out at 120 °C with 6 h. The solid substrates (solid yield) and delignification after DESs treatment of poplar and pine is presented in Fig. 5. For poplar, the delignification of LA-ATMAC-2:1 could reach 70.0%, which was slightly higher than that of LA-ChCl-2:1 (69.8%). For pine, 82.0% lignin in LA-ATMAC-2:1 was removed under the same conditions, however, only 54.2% lignin in LA-ChCl-2:1 was removed. Moreover, the two DESs have good removal ability for hemicellulose. These results show that LA-DESs have a unique ability to selectively extract lignin and hemicellulose while preserving the cellulose. In particular, LA-ATMAC-2:1 has excellent delignification ability. Table 4 shows that the delignification efficiency of the same DES on different raw materials is different (Table 4, Entry 2, 3, 13, 14). The delignification efficiency of LA-DES is higher than other acidic DESs (Table 4, Entry 3, 4, 8, 9, 10). And the lignin removal rate of raw materials increases with the increase of the molar ratio of acid in DES (Table 4, Entry 3, 5, 6). The -COOH groups in DESs provided active protons and facilitated the proton-catalyzed cleavage of lignin-

Table 3
The composition of two raw materials of lignocellulosic biomass.[a]

Raw material	AIL[b] (wt %)	ASL[c] (wt %)	Cellulose (wt %)	Hemicellulose (wt %)
Poplar	24.8	3.3	45.7	16.4
Pine	29.4	1.6	41.7	19.5

[a] The compositions of biomass were analyzed according to the procedures of the NREL method (Sluiter et al., 2010)
[b] AIL, acid insoluble lignin (klason lignin).
[c] ASL, acid-soluble lignin.

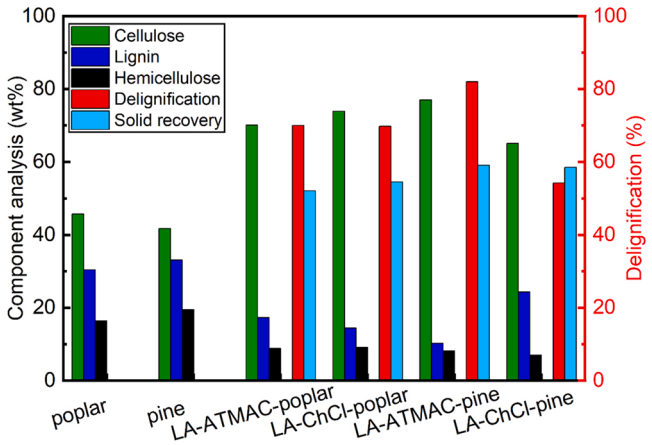


Fig. 5. Component analysis of raw materials, the solid substrates and delignification after DESs (molar ratio with 2:1) pretreatment.

carbohydrate complexes (LCCs) and ether and ester bonds in lignin (Lee et al., 2021). The results showed that LA-DESs have excellent lignin dissolution and delignification ability, especially for LA-ATMAC DES. Furthermore, it was also demonstrated that DES is a non-toxic, easy--to-prepare green solvent and can replace traditional solvents for lignin dissolution and extraction efficiently compared to traditional methods (sodium hydroxide, sulfate and sulfites and organic solvents).

3.5. Mechanism of KL dissolution and biomass pretreatment in the DESs

The dissolution of KL in the DESs could be explained from three aspects, including the solubility of KL, the K-T parameters of DESs, and the structural changes of regenerated KL. The functional groups in the HBAs and HBDs have significant effect on the dissolution of KL in DESs. The -COOH groups in LA and GA could provide active protons, which enhance the H-bonds strength and enable the cleavage of various chemical bonds in lignin (Zhou et al., 2022). However, the excess -COOH groups in HBDs (*e.g.*, MA) could form more H-bonds and extensive dimer chains, resulting in higher viscosity of DESs, which reduced the mass transfer rate of the solvent system. Hence, additional energy was required to loosen the bonded structures to initiate interactions with lignin. Furthermore, the presence of an extensive H-bonds network weakened the ability of HBAs to compete with intramolecular bonding in lignin, thus resulting in lower solubility of lignin. Moreover, for the DESs with the same HBD, the solubility of KL changed with the functional groups of the HBAs following the order: C═C > -OH > alkyl, which was consistent with the previously reported results (Liu et al., 2019b). The KL solubility showed good positive correlation with the values of the K-T parameters of the prepared DESs (α and π *) . DESs with a high α value could easily donate H-bonds to release H^+, hence a higher degree of lignin dissolution, suggesting that the dissolution of KL was mainly affected by the HBDs of the obtained DESs (Lee et al., 2021; Tan et al., 2019). As characterized by ^{1}H NMR, 2D HSQC NMR and FT-IR spectra, the dissolution and regeneration process involved the breaking of β-O-4, β-5 and β-β linkages, the main mechanism is as follows: DES provides active protons, and lignin is acylated in DES and positive protons are formed at Cα. The deprotonation and protonation processes are followed by the formation of a resonant stable positive charge, followed by dehydration and further fracture of the ether bond (Fig. 6), (Chen et al., 2020a; Xia et al., 2018) indicating that chemical reactions occurred when the KL was dissolved in DESs. Meanwhile, the decreased molecular weight and the lower decomposition temperature for the regenerated KL implied that the KL partially degraded during the dissolution process. Similarly, the -COOH groups in DESs could provide active protons during pretreatment of biomass. These active protons facilitate the proton-catalyzed cleavage of lignin-carbohydrate complexes (LCCs) and various chemical bonds (*e.g.*, ether and carbon-carbon bonds) in lignin.

4. Conclusions

In summary, DESs composed by carboxylic acid (*i.e.*, LA, MA, and GA) as the HBD and quaternary ammonium salt (*i.e.*, ATMAC, ChCl, TMAC, and TEAC) as the HBA, were designed for the dissolution of KL and pretreatment of poplar and pine. This study confirmed that the ability of DESs for KL dissolution is significantly affected by the chemical structure of HBDs and HBAs. The -COOH groups in HBDs could provide active protons, which enhance the H-bonds strength and enable the cleavage of various chemical bonds in lignin. However, the excess -COOH groups in HBDs could form more H-bonds and extensive dimer chains, resulting in higher viscosity of DESs and reduced the lignin dissolution and extraction during biomass pretreatment. Meanwhile, the functional group in the HBA affected the solubility of KL in DESs, following the order: C═C > -OH > alkyl group. The synergistic effect of LA and ATMAC afforded LA-ATMAC-2:1 the best developed DES in this work on KL dissolution (solubility of 22.5 wt% at 70 ℃) and lignocellulose delignification. Moreover, the solubility of KL shows good positive correlation with the K-T parameters α and π * of the developed DESs, while the correlation between KL solubility and β was poor. In addition, the content of β-O-4 linkages decreased after the KL was regenerated, indicating that DES achieved dissolution and extraction by breaking ether bonds during dissolution and pretreatment. DESs are a promising class of functional solvents to pretreat and dissolve lignin and even lignocellulosic biomass, and novel DES with high efficiency to pretreat biomass can be designed by the incorporating of proper HBDs and HBAs.

Table 4
Effects of DESs and conditions on the delignification of different raw materials.

Entry	Solvent	Molar ratio	Raw materials	Conditions	Delignification	Ref.
1	LA:ChCl	2:1	Wheat Straw	120 ℃/24 h	85.9%	(Lou et al., 2019)
2	LA:ChCl	2:1	Wheat Straw	120 ℃/6 h	55%	(Lou et al., 2019)
3	LA:ChCl	2:1	Corncob	90 ℃/24 h	64.7%	(Zhang et al., 2016)
4	GA:ChCl	2:1	Corncob	90 ℃/24 h	56.4%	(Zhang et al., 2016)
5	LA:ChCl	5:1	Corncob	90 ℃/24 h	77.9%	(Zhang et al., 2016)
6	LA:ChCl	10:1	Corncob	90 ℃/24 h	86.1%	(Zhang et al., 2016)
7	LA:Alanine	9:1	Wheat Straw	60 ℃/24 h	23.7%	(Jablonsky et al., 2019)
8	LA:ChCl	1:1	Empty Fruit Bunch	120 ℃/8 h	33.5%	(Tan et al., 2019)
9	MA:ChCl	1:1	Empty Fruit Bunch	120 ℃/8 h	22.9%	(Tan et al., 2019)
10	LeA:ChCl	2:1	Corncob	90 ℃/24 h	43.0%	(Zhang et al., 2016)
11	LA:ATMAC	2:1	Poplar	120 ℃/6 h	70.0%	This work
12	LA:ATMAC	2:1	Pine	120 ℃/6 h	82.0%	This work
13	LA:ChCl	2:1	Poplar	120 ℃/6 h	69.8%	This work
14	LA:ChCl	2:1	Pine	120 ℃/6 h	54.2%	This work

LA, GA, MA, and LeA represent lactic acid, glycolic acid, malic acid, and levulinic acid, respectively.

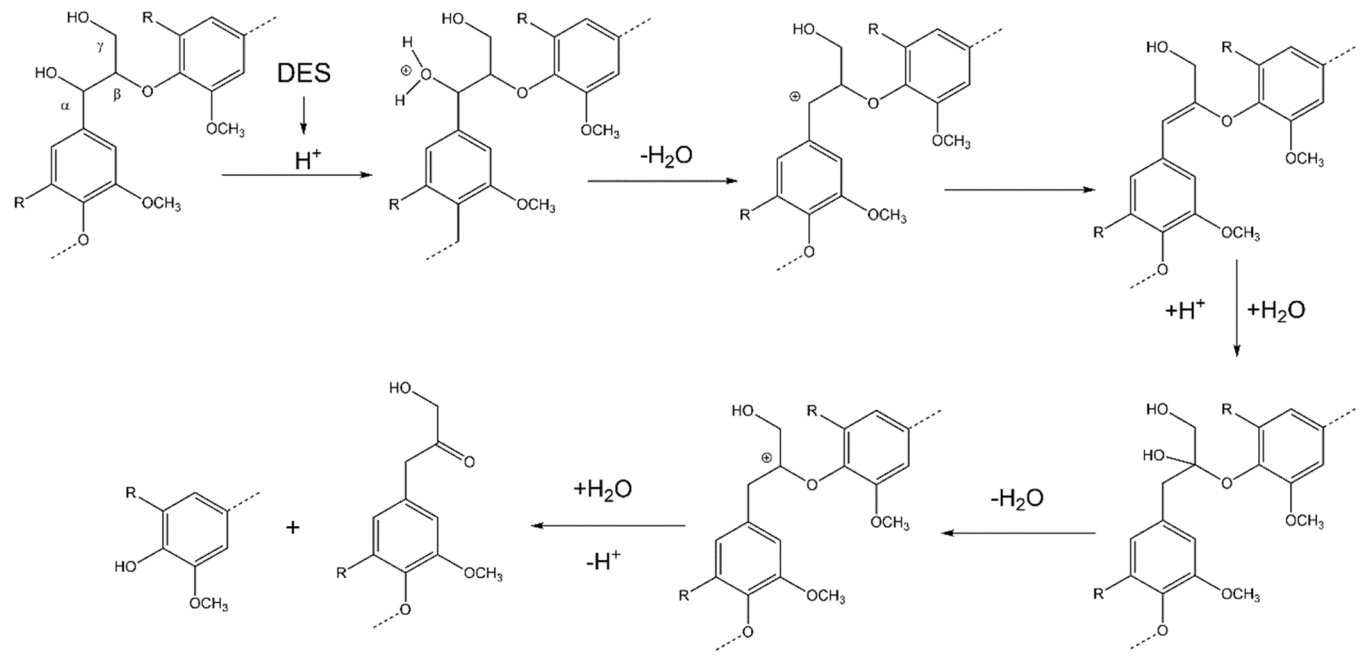


Fig. 6. Possible routes of the cleavage of β-O-4′ linkages during lignin dissolution and DES pretreatment.

CRediT authorship contribution statement

Haitao Yu: Investigation, Writing – original draft. **Zhimin Xue**: Conceptualization, Methodology, Writing – review & editing, Supervision. **Ruifen Shi**: Investigation. **Fengyi Zhou**: Investigation. **Tiancheng Mu**: Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhimin Xue and Tiancheng Mu reports financial support was provided by National Natural Science Foundation of China.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2022.115049.

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